
The Hallucination Trilemma: A Topological Impossibility for Faithful, Covering, and Calibrated Language Models

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Abstract

We prove that no question-answering model with a continuous confidence map on a connected question space can be simultaneously *faithful* (confidence $\geq \frac{1}{2}$ implies a correct answer), *covering* (it returns both correct and wrong answers), and *strictly calibrated* (confidence above/below $\frac{1}{2}$ iff correct/wrong). Coverage supplies witnesses on both sides of the threshold; connectedness and continuity force a threshold-confidence question through the intermediate value theorem; calibration pins its truth-distance to zero; faithfulness demands that it be strictly negative. The contradiction is structural—an application of the IVT—not a statistical claim about any current system. Crucially, the three conditions are not an adversarial construction but the ideal that model builders actively strive for: a model that is trustworthy when confident, useful across questions, and honest about what it knows. Our result shows this sought-after ideal is not merely hard to reach but formally unreachable, and the closer a model is driven toward it—as calibration is idealized toward the zero-slack biconditional—the tighter the obstruction binds. The same two-line core fires with no topology once a single boundary question exists, and the result extends to multi-turn, stochastic, and approximately calibrated settings, where it becomes quantitative. All results are machine-verified in Lean 4 against Mathlib v4.28, sorry-free, depending only on the three standard kernel axioms.

1 Introduction

Hallucination—a model producing a confident but wrong answer—is one of the most reliably observed failures of deployed language models (Maynez et al., 2020; Ji et al., 2023; Zhang et al., 2023). The usual fixes are empirical: more data, RLHF, temperature tuning, post-hoc calibration. We show that under idealized conditions there is a mathematical obstruction that no amount of training removes.

Consider any model that returns an answer together with a confidence score. Three properties are natural to want:

- **Faithful:** when at least half-confident, the answer is correct.
- **Covering:** the model is sometimes right and sometimes wrong—it does not refuse everything or output a constant.
- **Calibrated:** confidence tracks correctness in both directions—high confidence means right, low confidence means wrong.

For a continuous confidence map on a connected question space these three cannot coexist. The proof needs no training assumptions, no architecture, no statistics: it is the intermediate value theorem.

Why. Coverage gives a question q_+ answered correctly and a question q_- answered wrongly. Calibration turns these into confidence above $\frac{1}{2}$ at q_+ and below $\frac{1}{2}$ at q_- . If Q is connected and the confidence map is continuous, the IVT forces confidence to equal $\frac{1}{2}$ at some q_0 . Calibration then pins the answer at q_0 exactly

on the truth boundary. But faithfulness says half-confident answers must be strictly correct. At q_0 the model is exactly half-confident and its answer is on the boundary, not strictly correct—a contradiction.

What it says about real models. Coverage holds for every useful model; the binding condition is calibration. The theorem does *not* claim deployed LLMs meet all three and derive a contradiction—no current model satisfies strict biconditional calibration. It says the design target the field is working toward is unreachable, and that the constraint tightens as calibration improves. Section 4 makes this quantitative: under ε -calibration a threshold-confidence question has truth-distance in $[-\varepsilon, 0)$, collapsing to $0 \leq \delta_M(q_0) < 0$ at $\varepsilon = 0$.

Relation to prior work. Xu et al. (2024) prove hallucination via computability (finite capacity cannot cover all computable functions), an argument about the distribution boundary that ignores calibration. Karpowicz (2025) derive a conflict among four mechanism-design conditions via proper scoring rules. Neither uses the IVT or makes calibration central. Our result differs on all three counts—three conditions with calibration explicit, an IVT-only proof, and machine verification—and is, to our knowledge, the first hallucination impossibility formalized in a proof assistant.

2 Setup

Let Q be the *question space*, a connected topological space, and A the *answer space*. A *model* is a continuous map $M: Q \rightarrow A \times \mathbb{R}$ with components $\text{ans}(q) = (M(q))_1$ and $\text{conf}(q) = (M(q))_2$. Connectedness is the key hypothesis; the Lean proof uses only preconnectedness of Q and continuity of the confidence coordinate, not a path between questions.

Correctness is a continuous signed *truth-distance* $\delta: Q \times A \rightarrow \mathbb{R}$: $\delta(q, a) < 0$ means a is correct at q , $\delta(q, a) > 0$ means wrong, and $\delta(q, a) = 0$ is the boundary. Write $\delta_M(q) = \delta(q, \text{ans}(q))$. This subsumes soft factuality scores, semantic-similarity thresholds, and any score with a natural zero. The threshold $\frac{1}{2}$ is conventional; replacing it by any constant c gives an identical proof.

Definition 2.1 (Faithfulness). $\text{conf}(q) \geq \frac{1}{2} \Rightarrow \delta_M(q) < 0$ for every q .

Definition 2.2 (Coverage). *There exist q_+, q_- with $\delta_M(q_+) < 0$ and $\delta_M(q_-) > 0$.*

Definition 2.3 (Strict calibration). *For every q : $\text{conf}(q) > \frac{1}{2} \iff \delta_M(q) < 0$ and $\text{conf}(q) < \frac{1}{2} \iff \delta_M(q) > 0$.*

Strict calibration is a biconditional between the side of the threshold and the sign of the truth-distance. It differs from statistical calibration (where conf approximates the probability of correctness) and is the condition that gives the argument its bite: it pins δ_M to zero wherever confidence crosses $\frac{1}{2}$.

3 The Trilemma

Theorem 3.1 (Boundary question). *Let Q be connected and M continuous with continuous δ . If M is covering and strictly calibrated, there is $q_0 \in Q$ with $\text{conf}(q_0) = \frac{1}{2}$ and $\delta_M(q_0) = 0$.*

Proof. Coverage and calibration give $\text{conf}(q_+) > \frac{1}{2}$ and $\text{conf}(q_-) < \frac{1}{2}$. The confidence map is continuous on the connected Q , so by the IVT it equals $\frac{1}{2}$ at some q_0 . There it is neither above nor below the threshold, so both biconditionals fire negatively: $\delta_M(q_0)$ is neither < 0 nor > 0 , hence $= 0$. $\square$

Theorem 3.2 (Hallucination Trilemma). *No continuous model on a connected question space is simultaneously faithful, covering, and strictly calibrated.*

Proof. Theorem 3.1 gives q_0 with $\text{conf}(q_0) = \frac{1}{2}$ and $\delta_M(q_0) = 0$. Since $\text{conf}(q_0) \geq \frac{1}{2}$, faithfulness requires $\delta_M(q_0) < 0$, contradicting $\delta_M(q_0) = 0$. $\square$

The proof uses nothing about architecture, training, or parameter count—only continuity of M and connectedness of Q . Any model satisfying the three conditions must violate one of these two.

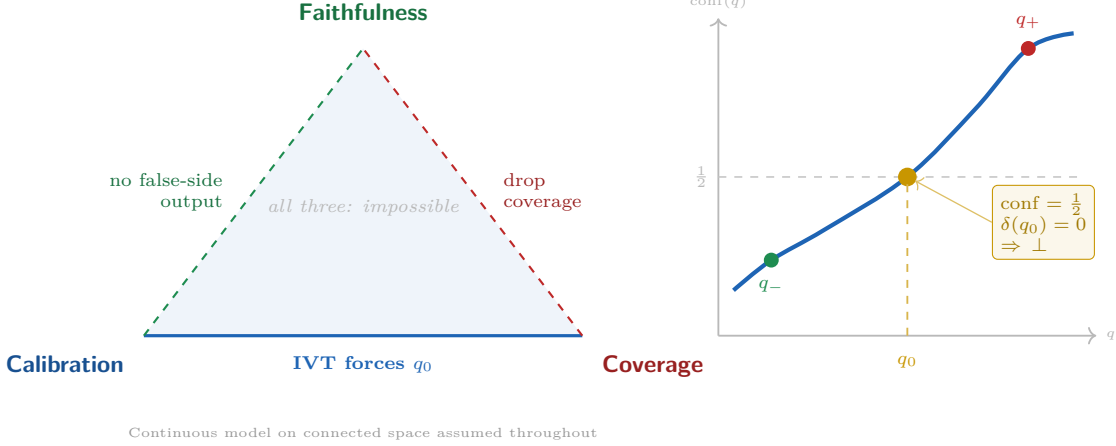


Figure 1: **Left:** the trilemma. No continuous model on a connected space satisfies all three; the solid edge is the main result, the dashed edges the counterexamples. **Right:** the IVT argument. Coverage gives q_- ($\text{conf} < \frac{1}{2}$) and q_+ ($\text{conf} > \frac{1}{2}$); connectedness forces the crossing q_0 ; calibration pins $\delta(q_0) = 0$; faithfulness then demands $\delta(q_0) < 0$ —contradiction.

Tightness. The threshold is exact: relaxing faithfulness by any $\varepsilon > 0$ breaks the impossibility. On $Q = [0, 1]$ with $\text{conf}(q) = 1 - q$ and $\delta_M(q) = q - \frac{1}{2}$, strict calibration and coverage hold, and $(\frac{1}{2} + \varepsilon)$ -faithfulness holds for every $\varepsilon > 0$; only standard faithfulness fails, at $q = \frac{1}{2}$. The band $[\frac{1}{2}, \frac{1}{2} + \varepsilon)$ is the price of escape.

Each hypothesis is necessary. Dropping any single condition lets the other two coexist (the dashed edges of Figure 1, left). *Not faithful:* the tightness model above—calibrated and covering, but $\delta_M = 0$ at $\text{conf} = \frac{1}{2}$. *Not covering:* constant $\text{conf} = \frac{3}{4}$ with $\delta_M < 0$ everywhere is faithful and calibrated, but has no wrong-answer witness. *Not continuous:* $\text{conf} = \frac{3}{4}$ on $(\frac{1}{2}, 1]$ and $\frac{1}{4}$ on $[0, \frac{1}{2})$ with matching signs is faithful, covering, and calibrated pointwise, but jumps at $\frac{1}{2}$.

4 Extensions

Discrete core. The IVT is used only to *produce* the boundary question; the contradiction itself needs no topology.

Theorem 4.1 (Discrete impossibility). *Let Q, A be arbitrary sets with no topology. No model can be faithful, strictly calibrated, and have a question q^* with $\delta_M(q^*) = 0$.*

Indeed, $\delta_M(q^*) = 0$ makes both calibration biconditionals fail unless $\text{conf}(q^*) = \frac{1}{2}$, and faithfulness then demands $\delta_M(q^*) < 0$. The continuous trilemma factors through this: IVT supplies q^* from two-sided coverage and connectedness (Figure 2b). On a genuinely disconnected set with no boundary question, all three conditions can hold—a real escape—so the discrete version assumes a boundary question (“three-sided coverage”). Equivalently: under strict calibration, a model is faithful iff it has *no* boundary questions.

Stochastic models. Randomizing at q_0 does not help. Let $\bar{c}, \bar{\delta}$ be expected confidence and truth-distance; if they are continuous and meet the three conditions in expectation, the trilemma yields q_0 with $\bar{\delta}(q_0) = 0$. The sample-level version is sharper.

Theorem 4.2 (Stochastic dichotomy). *Let $c, d: \Omega \rightarrow \mathbb{R}$ be measurable on a probability space (Ω, μ) with d integrable, $\int d d\mu = 0$, sample faithfulness ($c \geq \frac{1}{2} \Rightarrow d < 0$) μ -a.e., and $\mu\{c \geq \frac{1}{2}\} > 0$. Then $\mu\{d > 0\} > 0$.*

Proof. If $\mu\{d > 0\} = 0$ then $-d \geq 0$ a.e. with $\int (-d) d\mu = 0$, so $d = 0$ a.e. But on $\{c \geq \frac{1}{2}\}$ (positive probability) faithfulness gives $d < 0$ a.e.—a contradiction. $\square$

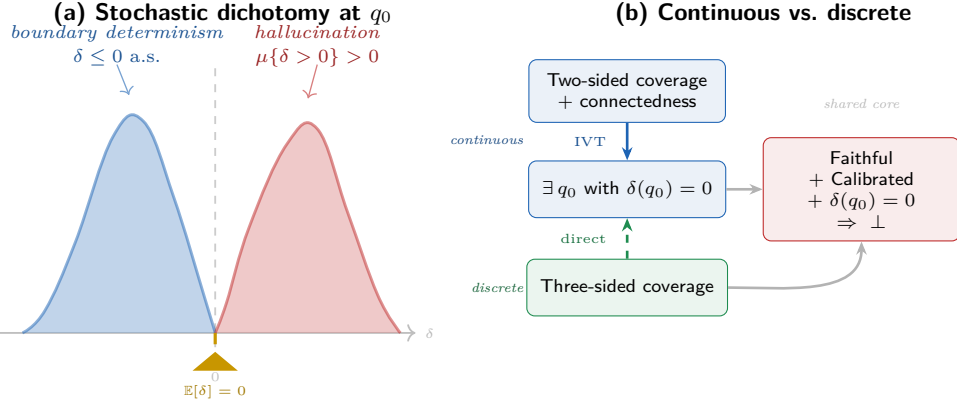


Figure 2: **(a)** At the forced point q_0 where $\mathbb{E}[\delta] = 0$, a confident stochastic model faces a dichotomy: all mass at $\delta = 0$ (boundary determinism) or positive mass at $\delta > 0$ (hallucination). **(b)** The continuous and discrete trilemmas share a two-line core (right), differing only in how the witness q_0 is produced—IVT from two-sided coverage vs. explicit three-sided coverage.

So a stochastic model that is confident with positive probability at q_0 hallucinates there with positive probability (Figure 2a).

Multi-turn and antipodal. A finite product of connected spaces is connected, so the history space Q^T is connected and Theorem 3.2 applies to history-dependent $M: Q^T \rightarrow A \times \mathbb{R}$ unchanged: context does not remove the obstruction. When questions come in opposite pairs via a continuous involution σ with $\delta_M(\sigma(q)) = -\delta_M(q)$, the IVT forces a zero with no coverage hypothesis at all—coverage becomes automatic.

Approximate calibration. Real models are not strictly calibrated. Call M ε -calibrated if $\delta_M(q) < -\varepsilon \Rightarrow \text{conf}(q) > c + \varepsilon$ and $\delta_M(q) > \varepsilon \Rightarrow \text{conf}(q) < c - \varepsilon$.

Theorem 4.3 (Approximate trilemma). *Let $\varepsilon \geq 0$, Q connected, conf continuous. If M has ε -coverage, is two-sided ε -calibrated, and is c -faithful, then some q_0 has $\text{conf}(q_0) = c$ and $\delta_M(q_0) \in [-\varepsilon, 0)$.*

Proof. ε -calibration sends the coverage witnesses to $\text{conf} > c$ and $\text{conf} < c$, so IVT gives $\text{conf}(q_0) = c$. Faithfulness gives $\delta_M(q_0) < 0$; and $\delta_M(q_0) < -\varepsilon$ would force $\text{conf}(q_0) > c + \varepsilon$, so $\delta_M(q_0) \geq -\varepsilon$. $\square$

At $\varepsilon = 0$ the conclusion is $0 \leq \delta_M(q_0) < 0$ —the exact contradiction. Improving calibration shrinks the margin: calibration and faithfulness are not independent. Taking Q_{in} correct and Q_{ood} wrong as the two witnesses, q_0 lies in neither, i.e. in the in-distribution/OOD transition band—the impossibility localizes there.

5 Formalization, Related Work, and Conclusion

Lean. All results are machine-verified in Lean 4 against Mathlib v4.28, with no `sorry` and every headline theorem reducing to `{propext, Quot.sound, Classical.choice}`. The core (HoF_07) applies `IsPreconnected.intermediate_value2` for the IVT step and two `linarith` calls for the calibration squeeze; the discrete, stochastic, and approximate results appear in HoF_10–12 (`integral_eq_zero_iff_of_nonneg_ae` for Theorem 4.2, `approx_trilemma` and `exact_from_approx` for Theorem 4.3). Sources are in the submission.

Related work. Empirical studies measure and taxonomize hallucination (Maynez et al., 2020; Ji et al., 2023; Zhang et al., 2023) without asking whether it is structurally unavoidable; the two prior impossibility results (Xu et al., 2024; Karpowicz, 2025) (Table 1) make calibration incidental rather than central. The calibration literature (Guo et al., 2017; Platt, 1999; Zadrozny & Elkan, 2001) treats miscalibration as correctable—the

Table 1: Comparison with prior hallucination impossibility results.

	Xu et al. (2024)	Karpowicz (2025)	This work
Proof technique	Computability	Mech. design + scoring	Topology (IVT)
Architecture-independent	✓	✓	✓
Addresses calibration	×	×	✓
Number of conditions	N/A	4	3
Machine-verified	×	×	✓

trilemma says correction costs one of the other two properties—while selective prediction (Kamath et al., 2020; Feng et al., 2020) is exactly dropping coverage. The companion Defense Trilemma (Anonymous, 2025) applies the same IVT engine to prompt-injection defenses; Lean has verified substantial mathematics before (Avigad, 2024; Buzzard et al., 2020), but this is the first such verification of a hallucination impossibility.

Conclusion. No continuous model on a connected question space is simultaneously faithful, covering, and strictly calibrated, by the intermediate value theorem alone. Context does not help while the history space stays connected, randomization gives positive-probability wrong answers, and discrete token space escapes only until a boundary question appears. The three conditions name the three available relaxations—abstain, tolerate a threshold band, or accept a weaker confidence signal—and at least one must give; which is cheapest depends on how often a deployment operates near its confidence boundary, an empirical quantity the theorem does not fix.

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